

1 INTRODUCTION

Online shopping generates large volumes of transactional data containing purchase time, product, quantity, price and customer information. Exploratory Data Analysis (EDA) helps turn these records into understandable patterns by revealing seasonality, high-demand products, customer concentration and geographic behaviour. These insights support better inventory planning, promotional timing and customer-retention decisions.

STUDY PERIOD
13 MONTHS
Dec 2010 – Dec 2011

RAW RECORDS
541,909
transaction lines

ATTRIBUTES
8 FIELDS
transaction + customer

PROBLEM STATEMENT

- Online sales patterns are difficult to prioritise from raw transaction logs.
- Seasonal peaks, product demand and repeat purchasing can be hidden in aggregate totals.

OBJECTIVES

- Analyse transaction and revenue patterns over time.
- Identify top products, peak shopping periods and customer behaviour.
- Translate EDA findings into practical e-commerce recommendations.

2 DATASET & STUDY DETAILS

DATASET
ONLINE RETAIL

REGION
UK-BASED
non-store retailer

VARIABLE
REVENUE
Quantity × UnitPrice

OBSERVATIONS
541,909
raw transaction lines

FIELDS
8
invoice to country

- InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country are the main fields.
- The source records transactions from 01/12/2010 to 09/12/2011 for a UK-based, registered non-store online retailer.
- Cancellation records are identifiable from invoice numbers beginning with “C”; cleaning decisions affect final totals.

3 ARCHITECTURE / METHODOLOGY

RAW DATA

UCI Online Retail

FILTER

Valid purchases

PREPROCESS

Missing & returns

FEATURES

Revenue + time

STATISTICS

Counts & totals

COMPARE

Trends & rankings

The workflow converts raw transaction records into an analysis-ready table, derives revenue and calendar features, then aggregates transactions by month, product, customer and geography. EDA is used to compare frequency, revenue and concentration before translating the strongest patterns into business actions.

4 KEY SHOPPING TRENDS IDENTIFIED

SEASONALITY

Q4 surge

Revenue rises sharply from September and peaks in November, indicating strong holiday-gifting demand.

PEAK SHOPPING TIME

12 PM

Public EDA summaries of the same dataset show midday as the busiest transaction window.

PRODUCT DEMAND

Giftware leads

Decorative gift items and themed products dominate high-volume and high-revenue rankings.

CUSTOMER BEHAVIOUR

Repeat buyers matter

A smaller group of frequent, high-value customers contributes disproportionately to sales.

5 RESULTS — ONLINE SHOPPING DATA ANALYSIS

RAW RECORDS
541,909
UCI dataset

CLEANED ROWS
406,829
common EDA subset

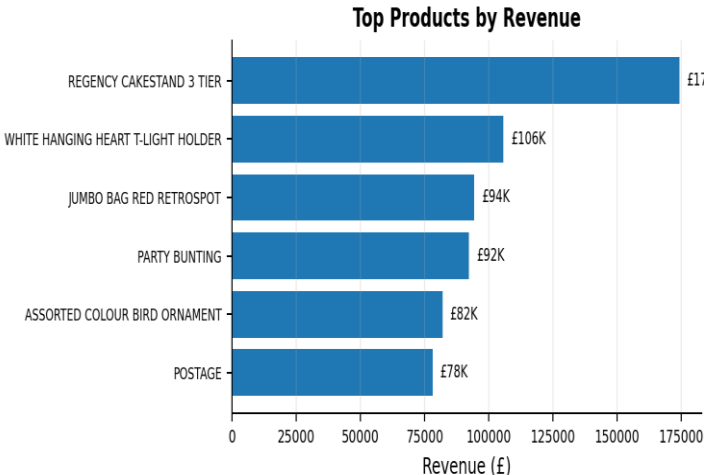
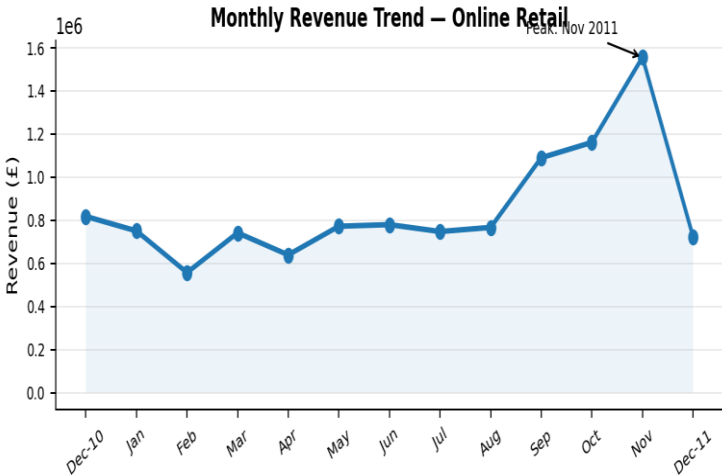
PEAK MONTH
NOV 2011
highest revenue

PEAK HOUR
12 PM
highest activity

TOP MARKET
UK
largest revenue base

MONTHLY REVENUE TREND

TOP PRODUCTS BY REVENUE



STATISTICAL RESULTS

- Raw records: 541,909
- Common cleaned analytical subset: 406,829
- Highest monthly revenue: November 2011
- Highest shopping activity: around 12 PM

KEY OBSERVATIONS

Holiday demand creates a strong late-year uplift. Decorative gift products repeatedly appear among top sellers, while customer activity is concentrated in a relatively small set of high-value buyers.

6 CONCLUSION & RECOMMENDATIONS

CONCLUSION

EDA of the Online Retail dataset shows that online shopping demand is highly seasonal, with a pronounced Q4 uplift and a November peak. Product-level rankings are dominated by gift-oriented merchandise, while transaction activity is concentrated around the daytime shopping window. These patterns show how EDA can connect raw e-commerce records with decisions on inventory, promotions and retention.

RECOMMENDATIONS

Seasonal inventory

Stock high-demand giftware ahead of the Sep–Nov surge.

Customer retention

Use RFM-style segments to reward frequent and high-value buyers.

Campaign timing

Concentrate promotions and remarketing before the November peak.

Product strategy

Bundle top gift items and test related-product cross-sells.